

Module 03: Dynamic Storage Allocation Strategies

1. Allocation Strategies Overview

Strategy	Algorithm Rule	Search Overhead	Key Advantage / Disadvantage
First Fit	Pehla hole jo size satisfy kare	Fast ($O(1)$ to $O(n)$)	Fastest; early memory fragmentation
Best Fit	Smallest hole jo size satisfy kare	Strictly $O(n)$	Preserves large holes; creates tiny useless slivers
Worst Fit	Largest available hole	Strictly $O(n)$	Large leftover; destroys largest blocks quickly
Next Fit	Last allocated pointer se circular First Fit	Fast	Spreads allocation evenly

Dynamic Storage Allocation Strategies: Comparison

1. First Fit

Working Principle:

Pehle hole se search shuru karo; jo PEHLA hole process size se bada mila, allocate kar do.

Advantages:

• Fastest execution speed

- Poori list search nahi karni padti
- Disadvantages:
- Shuruati memory blocks me tiny unusable holes jama ho jate hain.
- Time Complexity: $O(n)$ worst / $O(1)$ best
2. Best Fit
- Working Principle:
- Poori list search karo aur wo hole chuno jo process se bada ho aur SMALLEST ho.
- Advantages:
- Badi holes preserve rehti hain future ke bade processes ke liye.
- Disadvantages:
- Generates smallest leftover holes jo kisi kaam ke nahi hote (Worst fragmentation).
- Time Complexity: Strictly $O(n)$
3. Worst Fit
- Working Principle:
- Poori list me se LARGEST available hole ko choose karo aur allocate karo.
- Advantages:
- Leftover hole ka size bada hota hai, jisme naya process fit ho sakta hai.
- Disadvantages:
- System ke largest holes jaldi khatam ho jate hain; bade jobs starve karte hain.
- Time Complexity: Strictly $O(n)$
4. Next Fit & Compact Next Fit Strategy:
- First Fit ki tarah, lekin search start se nahi balki LAST ALLOCATION pointer se hoti hai.
- Compaction (Defra):
- External fragmentation khatam karne ke liye saare processes ko memory ke ek side shift karna taaki ek bada contiguous free block ban jaye.
- Compaction Condition:
- Possible ONLY if Dynamic Relocation is supported at runtime!

2. Solved Numerical Summary

Holes: 100K, 500K, 200K, 300K, 600K | **Processes:** P1(212K), P2(417K), P3(112K), P4(426K)

- **First Fit:** P1 in 500K, P2 in 600K, P3 in 500K remaining, **P4 Waits (Cannot Allocate).**
- **Best Fit:** P1 in 300K, P2 in 500K, P3 in 200K, P4 in 600K → **All 4 Processes Successfully Allocated!**
- **Worst Fit:** P1 in 600K, P2 in 500K, P3 in 600K remaining, **P4 Waits (Cannot Allocate).**

3. Compaction (Defragmentation)

External fragmentation ko khatam karne ke liye saare processes ko relocate karke contiguous hole banaya jata hai. Iske liye **Execution Time Dynamic Relocation** mandatory hota hai.